📏 Measurement Stations: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 1 · Level 3 · goes with Measurement Lab Sheet

Name: _____

Question: Are the station parts within the manufacturer's specification?

Materials

dial caliper, micrometer, feeler gauges, ruler, station parts, spec sheet

⚠️ Safety

- Safety glasses on
- Handle precision tools gently and return them to cases

Prediction

Which part do you predict will be most worn, and why?
Answers vary; crankpin journals often wear most.

Procedure

1. Clean each part.
2. Measure the rod diameter and length with the caliper.
3. Measure 3 crankshaft journals with the micrometer.
4. Find the feeler blade for the sample gap.
5. Record measured vs spec; decide in or out of spec.

Data

Part	Spec	Measured	In spec?
Rod diameter	**0.500**	**0.499**	**yes**
Magneto journal	**0.873**	**0.872**	**yes**
Crankpin	**0.873**	**0.868**	**no (worn)**
PTO journal	**1.001**	**1.000**	**yes**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: is this crankshaft reusable? Evidence: your measurements vs spec. Reasoning: what wear does to the engine.
Example: crankpin is 0.005 under spec, so it is worn; replace it or it will knock and lose oil pressure.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*